Computer Science & IT

COMPILER DESIGN
(CD)

Parsers

Lecture No. 13



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Pause

Topics to be Covered



- Topic
- Topic
- Topic

Parsers

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Inspiring Stories : Baby Halder

Baby Halder – The Maid Who Became a Bestselling Author

Background: Child bride at 12. Mother of 3 by 20. Worked as a domestic help in Delhi to survive.

Struggles: Abusive marriage, abandoned with children. Cleaned houses during the day, raised kids, wrote at night with discarded notebooks.

Achievements: Wrote "A Life Less Ordinary", a raw memoir translated into over 20 languages. Invited to global lit fests.

Impact: She didn't have peace, privilege, or polish — just persistence. If she could write her way out of despair, you can code your way out of confusion.



Inspiring Stories : Julián Carrillo

Julián Carrillo – The Man Who Heard What No One Else Did

Background: Mexican composer born in 1875. Grew up surrounded by poverty but obsessed with sound.

Struggles: Ridiculed for claiming there were microtones between piano notes. Science rejected him. He lost positions, funding, reputation.

Achievements: Developed the "13th Sound" theory, now foundational to microtonal music. His instruments are studied even today.

Impact: Everyone thought he was hearing wrong — he was just hearing ahead of his time. If he could persist through rejection, you can persist through revision.



Inspiring Stories : Kavita Thakur

Kavita Thakur – The Kabaddi Girl from the Hills

Background: Born in a remote Himachal village, daughter of a sweeper in a local school. No shoes, no mat, no coach.

Struggles: Trained barefoot on rocky terrain, faced taunts for playing a "boy's game." The family could barely afford to travel.

Achievements: Represented India in the 2014 Asian Games, brought home Gold in Kabaddi.

Impact: From dust to podium, without ever complaining. If she could wrestle out of poverty, you can wrestle out of low ranks.

Inspiring Stories : Rajesh Sharma

Rajesh Sharma (IIT Bombay – Computer Science)

Background: **Son of a carpenter from Jodhpur; suffered from a clubfoot and couldn't afford coaching.**

Education: **BTech from IIT Bombay, cleared JEE via self-study.**

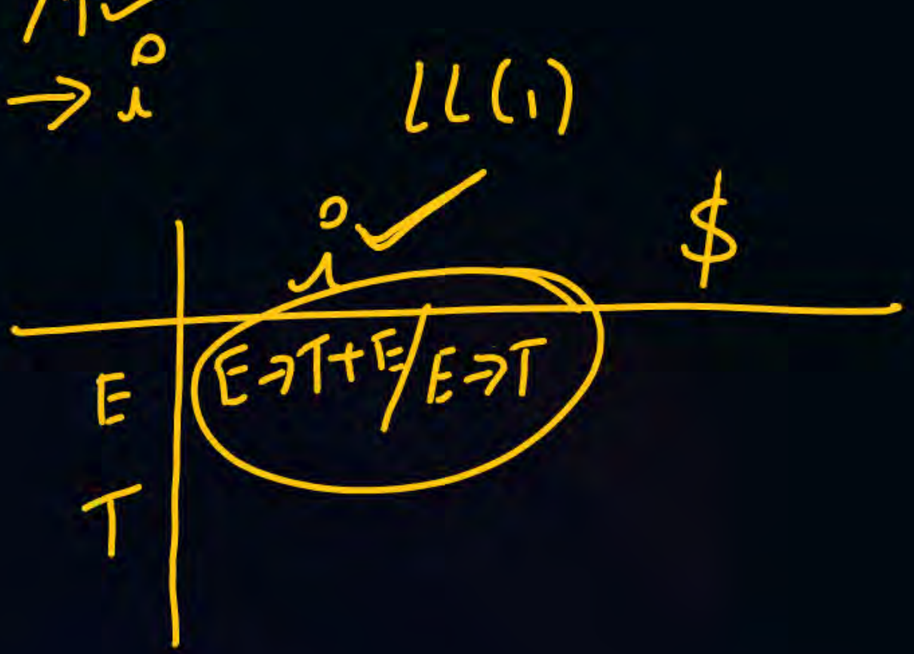
Career Achievements: **Secured a ₹32 LPA job at Rakuten in Japan during campus placements.**

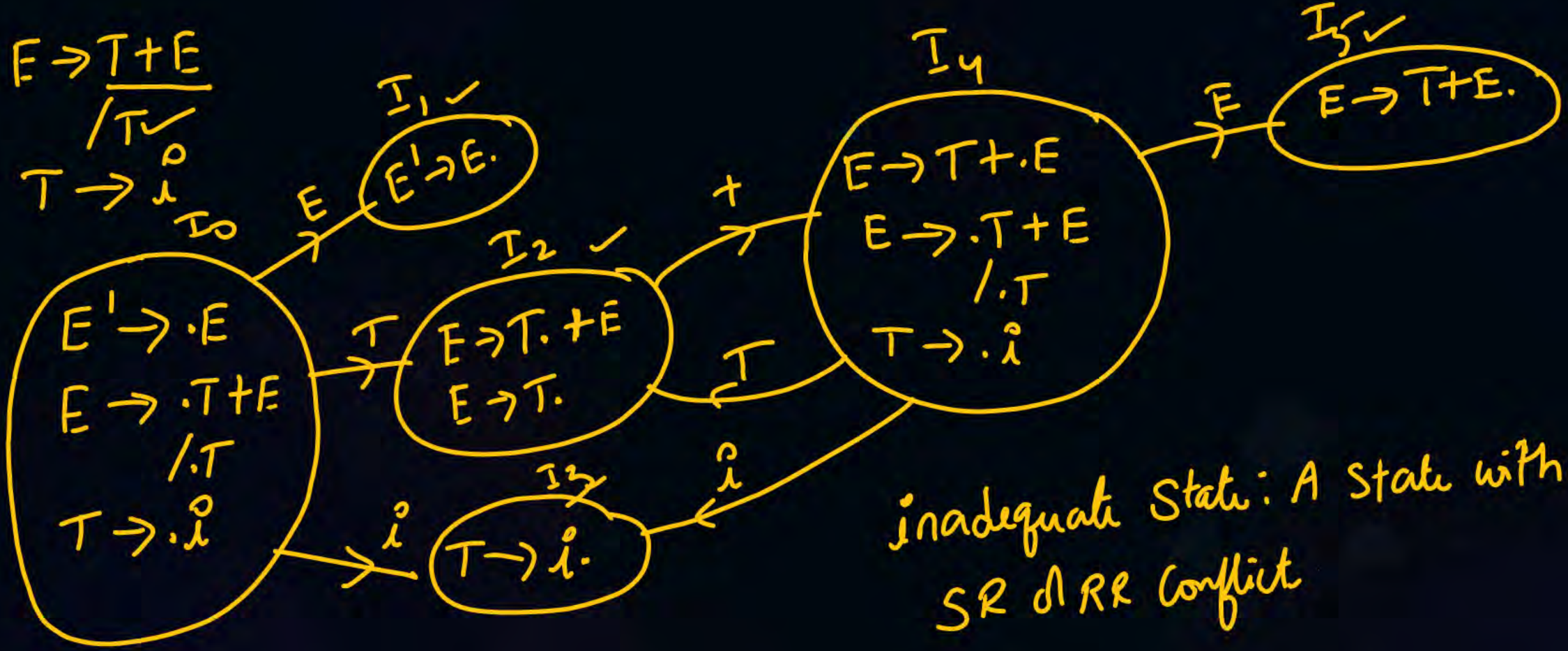
Impact: **Shattered stereotypes about disability and financial constraints, inspiring others to pursue big dreams.**

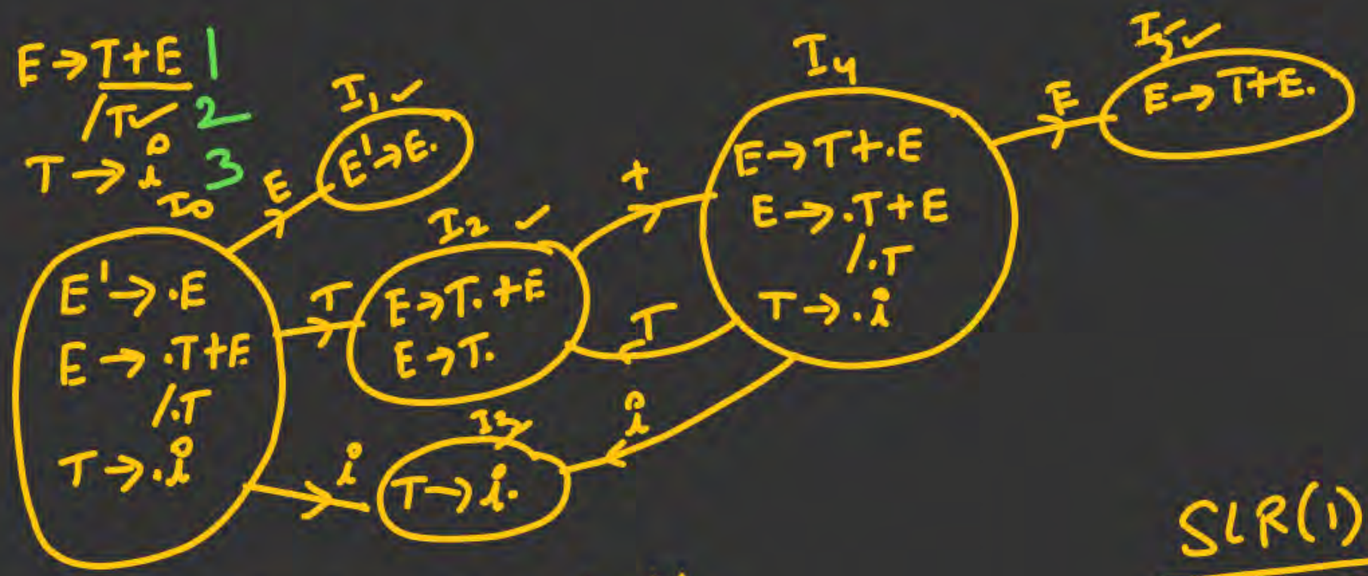
$E \rightarrow T+E$

$T \rightarrow i$

LL(1)




E→T+E
/T
T→i
I0
E'→.E
E→.T+E
/.T
T→.i
I1
E'→E.
I2
E→T.+E
E→T.
I3
T→i.
I4
E→T+.E
E→.T+E
/.T
T→.i
I5
E→T+E.
E
T
+
i
i
T
E
i
inadequate State: A State with either
SR or RR Conflict



Foll(E) = {$}

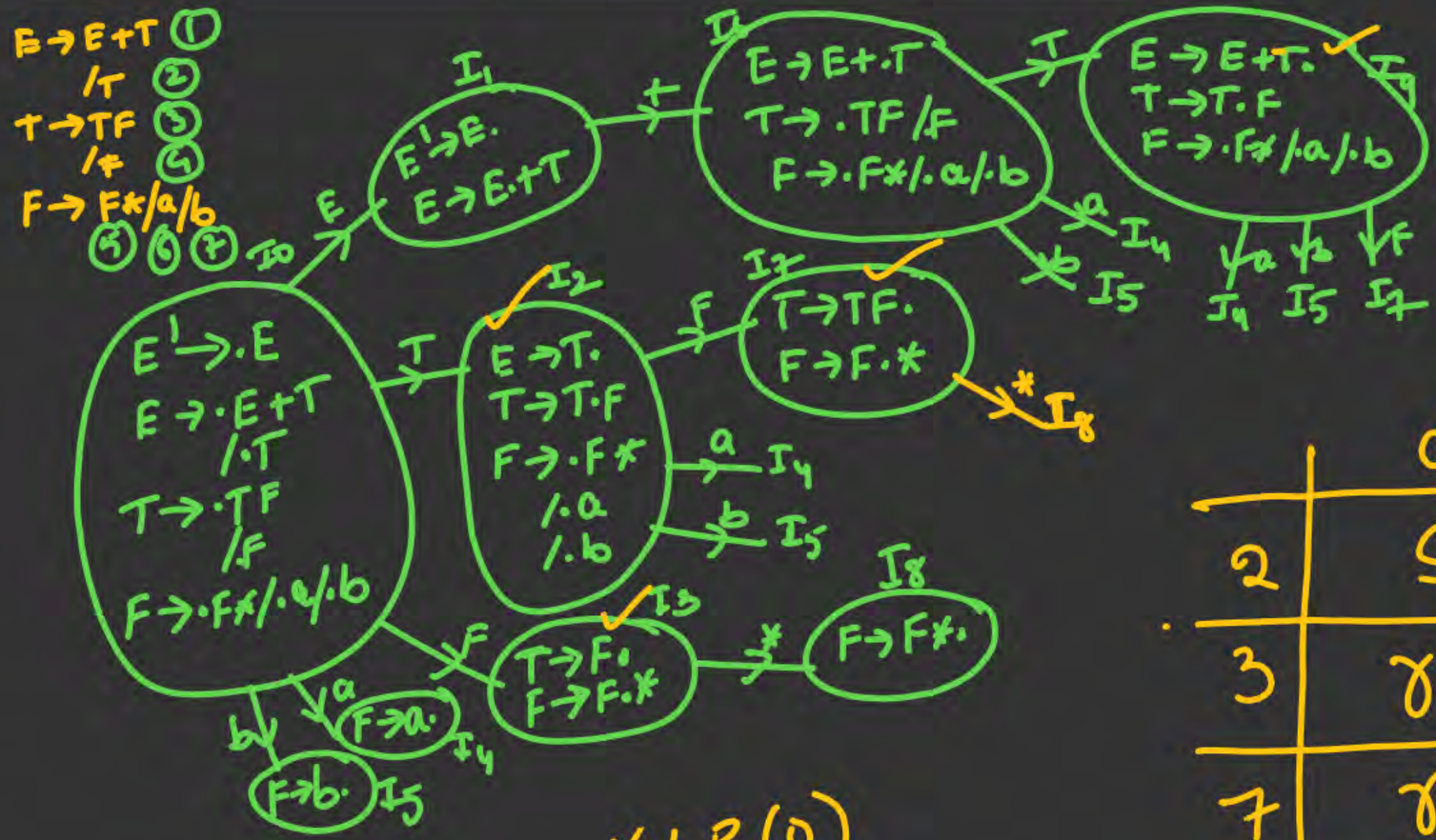
Foll(T) = {+, $}

LR(0) parsing table

	i	+	$	E	T
0	S_3			1	2
1			acc		
2	γ_2	γ_2/S_4	γ_2		
3	γ_3	γ_3	γ_3		
4	S_3			5	2
5	γ_1	γ_1	γ_1		

SLR(1) parsing table

	i	+	$	E	T
0	S_3			1	2
1			acc		
2		S_4	γ_2		
3		γ_3	γ_3		
4	S_3			5	2
5			γ_1		



XLR(0)

✓SLR(1)

Follow(E) = {+, $}

Follow(T) = {+, $, a, b}

Follow(F) = {*, +, $, a, b}

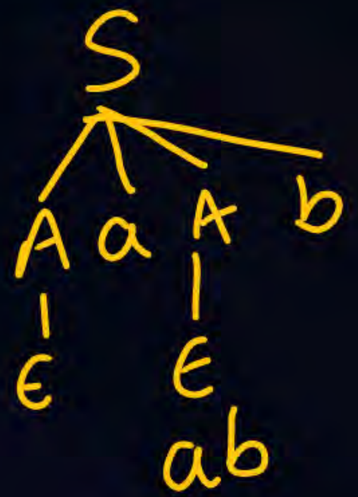
	a	b	+	*	$
2	S_4	S_5	γ_2		γ_2
3	γ_4	γ_4	γ_4	S_8	γ_4
7	γ_3	γ_3	γ_3	S_8	γ_3
a	S_4	S_5	γ_1		γ_1

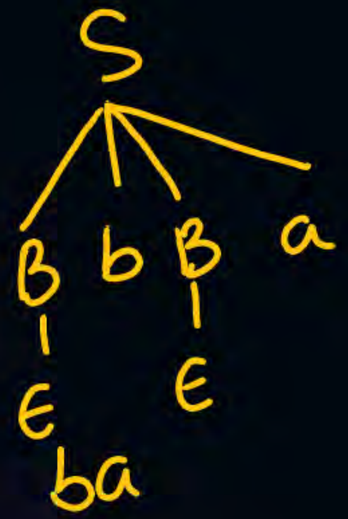
$S \rightarrow AaAb$

$\quad / BbBa$

$A \rightarrow \epsilon$

$B \rightarrow \epsilon$





$S \rightarrow AaAb$ ✓

$\quad / BbBa$ ✓

$A \rightarrow \epsilon$ ✓

$B \rightarrow \epsilon$ ✓

	a	b
S	$S \rightarrow AaAb$	$S \rightarrow BbBa$

$S \to AaAb$ ①

$/BbBa$ ②

$A \to \epsilon$ ③

$B \to \epsilon$ ④

I_0

$S' \to .S$

$S \to .AaAb$

$/.BbBa$

$A \to .$

$B \to .$

Not LR(0)

Not SLR(1)

	a	b	$
0	r_3/r_4	r_3/r_4	

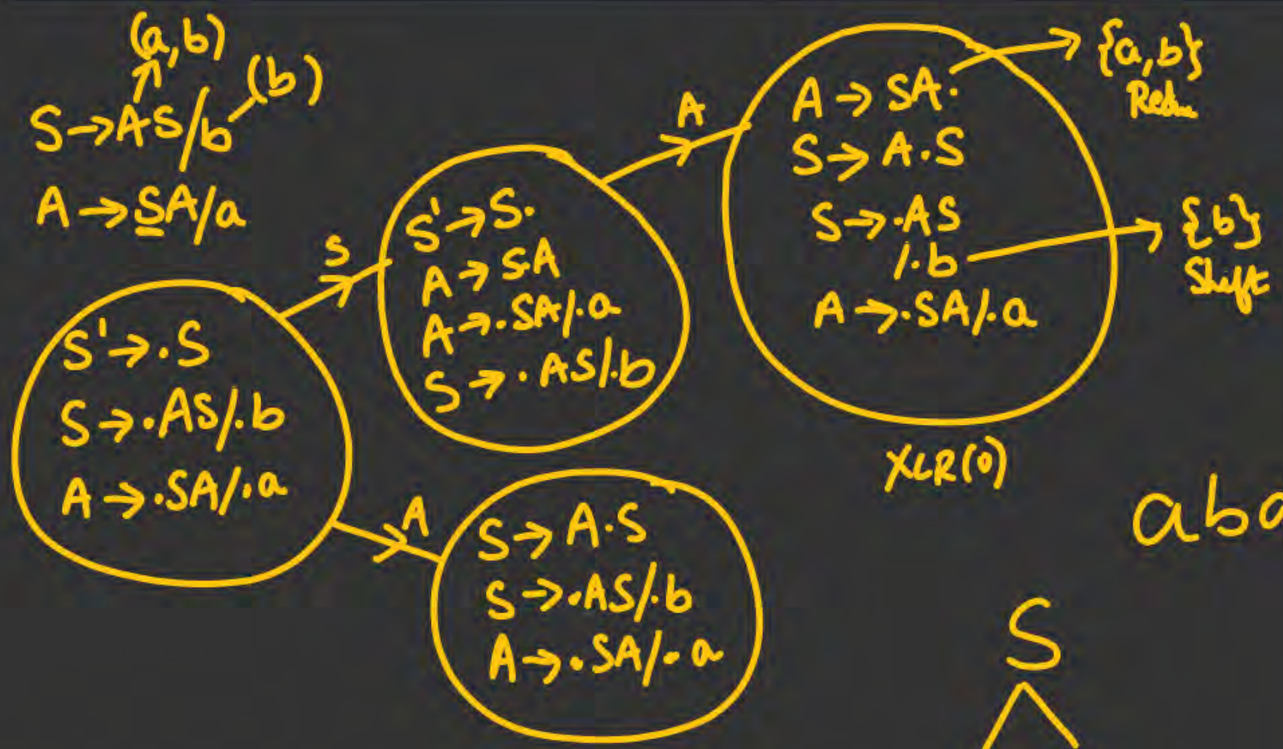
(a,b) (b)

$S \to AS / b$

$A \to SA / a$

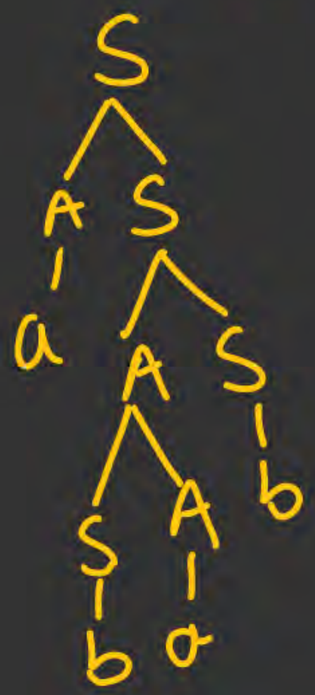
LL(1) ✗

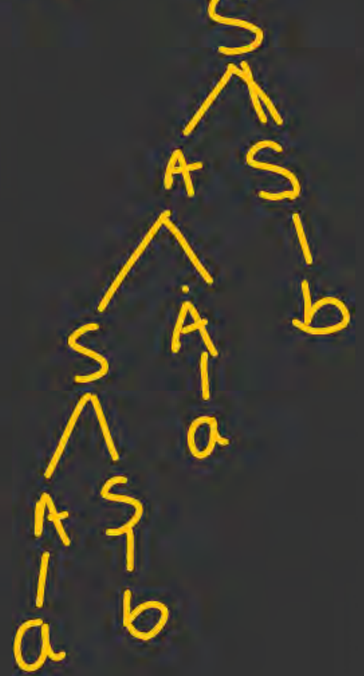
	b
S	$S \to AS / S \to b$



LL(1) ✗ SLR(1) ✗ LR(0)

abab





ambiguous

✗ LL(1)

$S \rightarrow Aa \quad \{d\}$

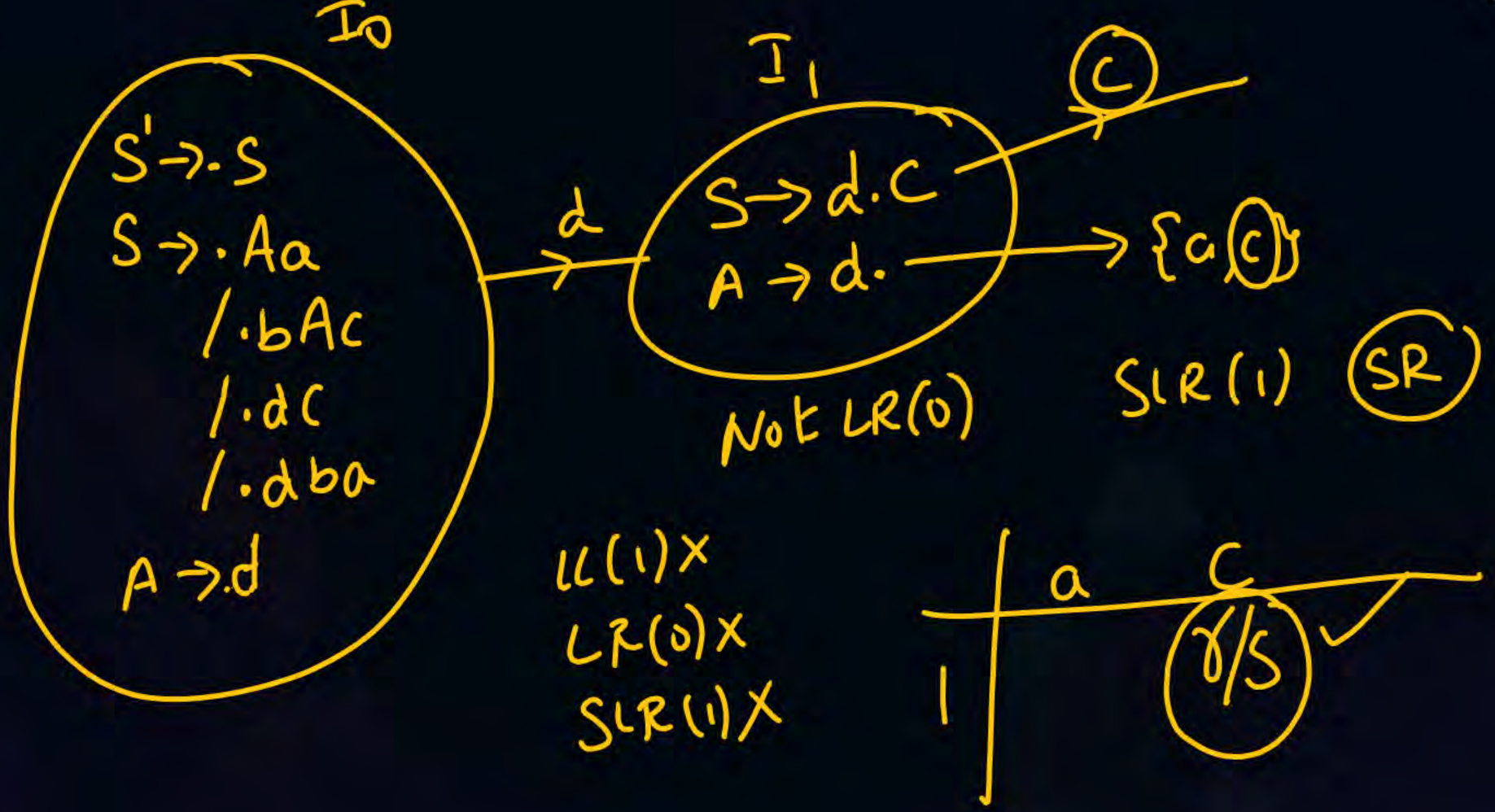
$/bAC$

$/dC \quad \{d\}$

$/bda$

$A \rightarrow d$

$S \to Aa$
$/bAC$
$/dC$
$/bda$
$A \to d$



Not LR(0)

SLR(1) (SR)

LL(1) X
LR(0) X
SLR(1) X

	a	c
1		γ/S

THANK - YOU